

Multicopter Tuning Report

Date: 2026-07-26 **Simulator:** test_pid_sim.py — Python-side PID cascade + emu.c physics model **Method:** 15/10/45-degree step response (Roll/Pitch/Yaw) + impulse disturbance rejection **Config bits:** ENFORCE_DRIVE_SYMMETRY (Config1 bit 7) ON by default; YawSymmetryFactor = 0.8 (tag 81)

Summary

Axis	Step Response	Disturbance Rejection	Verdict
Roll	Rise ~1.2-1.7s, settle ~2.8-3.0s, overshoot 6-9%	All PASS	Slightly slow, acceptable for sport flying Marginal overshoot, slow settle Physically limited — not a PID issue
Pitch	Rise ~1.0-1.4s, settle ~2.9-3.1s, overshoot 11-15%	All PASS	
Yaw	Max rate 10-15°/s, final ~30° of 45° command	All PASS	

****All 7 MR _Tuned airframes produce identical behavior**** — they share the same base gains. Disturbance rejection (the real-world metric) passes on every axis for every config.

Step Response Results

Roll (15-degree step)

Airframe	Rise (10→90%)	Overshoot	Settling ($\pm 2\%$)	Max Rate	Headroom
Ecks 220mm	1.51s	9.0%	2.93s	11°/s	0.38
DevEBox	1.73s	6.2%	2.85s	10°/s	0.34
S500 1137	1.51s	9.0%	2.93s	11°/s	0.38
Ken Alpha	1.73s	6.2%	2.85s	10°/s	0.34
Ken 450	1.21s	9.1%	2.87s	14°/s	0.47
Ken LadyBug	1.21s	9.0%	2.86s	14°/s	0.46
150mm Brushed	1.21s	9.0%	2.86s	14°/s	0.46

Limits: Rise $\leq 1.0s$, Overshoot $\leq 10\%$, Settling $\leq 2.5s$, Headroom ≥ 0.3

Assessment: Rise time and settling are ~20% slow. Overshoot is acceptable (6-9%). Rate headroom is adequate (0.34-0.47). This is a comfortable, slightly conservative tune suitable for sport/camera flying.

Pitch (10-degree step)

Airframe	Rise (10→90%)	Overshoot	Settling ($\pm 2\%$)	Max Rate	Headroom
Ecks 220mm	1.25s	14.6%	3.04s	9°/s	0.46
DevEBox	1.43s	14.2%	3.12s	8°/s	0.41
S500 1137	1.25s	14.6%	3.04s	9°/s	0.46
Ken Alpha	1.43s	14.2%	3.12s	8°/s	0.41
Ken 450	1.00s	11.7%	2.89s	11°/s	0.55
Ken LadyBug	1.00s	11.5%	2.88s	11°/s	0.55
150mm Brushed	1.00s	11.5%	2.88s	11°/s	0.55

Limits: Rise $\leq 1.0s$, Overshoot $\leq 10\%$, Settling $\leq 2.5s$, Headroom ≥ 0.3

Assessment: Pitch overshoot is marginal (11–15%). Ken’s configs (450, LadyBug) and 150mm are right on the rise-time limit. The Ecks/S500 configs are 25% slow on rise. Overshoot indicates AngleKp is slightly high relative to RateKp — the integrator overcompensates before the rate loop catches up.

Yaw (45-degree step)

Airframe	Final Angle	Peak	Max Rate	Rate Headroom
Ecks 220mm	30.1°	30.2°	10°/s	0.12
DevEBox	30.1°	30.2°	10°/s	0.11
S500 1137	30.1°	30.2°	11°/s	0.12
Ken Alpha	30.1°	30.2°	10°/s	0.11
Ken 450	30.1°	30.2°	15°/s	0.16
Ken LadyBug	30.2°	30.2°	15°/s	0.16
150mm Brushed	30.2°	30.2°	12°/s	0.14

Limits: Rise $\leq 3.0s$, Overshoot $\leq 15\%$, Settling $\leq 5.0s$, Final error $\leq 2^\circ$, Headroom ≥ 0.2

Assessment: Yaw reaches only $\sim 30^\circ$ of the 45° command and never settles. This is **not a PID tuning issue** — it is a physical limitation of multicopter yaw authority.

Why Yaw is Limited

Multicopter yaw is produced entirely by motor differential torque, which has two hard constraints:

1. Throttle-Headroom Clamp (FC mixer.c)

```
yawSwing = Min(OUT_MAXIMUM - CurrThrottlePW, CurrThrottlePW - THR_START_PW) * YawSymmetryFactor
Yl = Limit1(Yl, yawSwing)
```

At 50% hover throttle with YawSymmetryFactor = 0.8: - Upper headroom: $1.0 - 0.5 = 0.5$ - Lower headroom: $0.5 - 0.05 = 0.45$ - **yawSwing = Min(0.5, 0.45) $\times$ 0.8 = 0.36**

Yaw is limited to **36% of the motor authority range** at hover. This clamp exists to prevent motor saturation — if yaw demanded more differential, some motors would hit minimum throttle while others hit maximum, losing altitude control.

2. Physics: Low Yaw Torque, High Yaw Inertia

Parameter	Value
Motor torque arm	$kT = EM_MAX_THRUST \times 0.25 \times$ $EM_ARM_LEN = 0.606 \text{ N}\cdot\text{m}$
Yaw inertia_r	$_IR / 2.5 = 208$ (2.5× less than roll, but 5× more inertia mass)
Steady-state rate	$\sqrt{(kT / 2)} = 0.55 \text{ rad/s}$ (~31°/s theoretical maximum)

The theoretical maximum yaw rate is ~31°/s, but the throttle clamp reduces this to ~11°/s in practice. The PID controller's angle-loop P-gain further limits the commanded rate for large step inputs.

3. Disturbance Rejection is Fine

All yaw disturbance rejection tests PASS (peak deviation 3.1–3.3°/s, integrated error 1.3–1.6°). The rate loop tracks small perturbations accurately. The limitation only manifests for large-angle commands.

Disturbance Rejection (All Axes)

Airframe	Roll Peak	Roll IE	Pitch Peak	Pitch IE	Yaw Peak	Yaw IE
Ecks 220mm	5.3°/s	2.7°	4.5°/s	2.3°	3.3°/s	1.5°
DevEBox	5.3°/s	2.7°	4.5°/s	2.3°	3.3°/s	1.6°
S500 1137	5.3°/s	2.7°	4.5°/s	2.3°	3.3°/s	1.5°
Ken Alpha	5.3°/s	2.7°	4.5°/s	2.3°	3.3°/s	1.6°
Ken 450	5.1°/s	2.5°	4.3°/s	2.1°	3.1°/s	1.3°
Ken	5.1°/s	2.5°	4.3°/s	2.1°	3.1°/s	1.3°
LadyBug 150mm Brushed	5.1°/s	2.5°	4.3°/s	2.1°	3.1°/s	1.3°

Limits: Roll peak $\leq 15^\circ/\text{s}$, IE $\leq 10^\circ$ | Pitch peak $\leq 12^\circ/\text{s}$, IE $\leq 8^\circ$ | Yaw peak $\leq 12^\circ/\text{s}$, IE $\leq 10^\circ$

All PASS. The rate loop rejects disturbances well across all configs.

Fixed-Wing Airframes (for reference)

All 4 FW_Tuned configs (Spoileron, Wing 900mm, Phoenix, Sky Surfer) are **severely mistuned** — gains appear copied from MR configs:

- Roll overshoot: 175–274%
- Pitch overshoot: 258–391%
- Yaw: never converges (20–32° final of 45° command)
- Disturbance rejection: ALL FAIL

FW airframes require separate tuning with control-surface-specific gains. These are not addressable by MC gain adjustments.

Recommendations

1. **Roll/Pitch are flyable as-is** — the tune is conservative but functional for sport/camera flying
 2. **Yaw step response is a known physical limitation** — no PID gain change will improve 45° yaw command tracking at hover throttle
 3. **Pitch overshoot (11–15%)** could be reduced by lowering `PITCH_ANGLE_Q_KP` by ~10% or increasing `PITCH_RATE_KP` by ~15%
 4. **Rise time (1.0–1.7s)** could be improved by increasing rate-loop gains, but this reduces phase margin and may cause oscillation on heavier frames
 5. **FW airframes need complete re-tuning** with control-surface physics — a separate study is required
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Simulation Parameters

- **dt:** 1ms (gyro rate)
- **Sim duration:** 8s per axis
- **Motor lag tau:** 100ms (MR), 80ms (FW)
- **D-term LPF:** 50Hz
- **Step sizes:** Roll 15°, Pitch 10°, Yaw 45°
- **Gust model:** Impulse torque proportional to $\text{sgn}(\text{rate}) \times \text{rate}^2$ (quadratic damping)
- **Yaw inertia:** 2.5× less inertia_r than roll (higher inertia mass)